Duration: 2 hours

THE UNIVERSITY OF MANCHESTER

EART 22201 Igneous Minerals and Processes

Practical test: Practice paper

DATE

TIME

Answer ALL questions in the booklets provided.

Clearly label which question each of your answers corresponds to. All writing must be LEGIBLE. Illegible documents cannot be marked.

There are 100 marks available on this paper. Section A is worth 40 marks, Section B is worth 40 marks, and Section C is worth 20 marks. The number of marks for each individual question is shown in square brackets and may be used as a guide to how much time you should spend on each.

In the first 45 minutes of the test you will be provided with EITHER the thin sections for Section A OR the thin section for Section B. In the second 45 minutes of the test you will be provided with the alternate thin section(s).

In the final 30 minutes of the test you should complete Section C, and you may also request to view any of the thin sections for a second time.

You are encouraged to use labelled sketches and diagrams to illustrate your answers.

Time management is important in this test. Aim to budget about one minute per mark (100 minutes) – if you work to this timeframe you will have 20 minutes for checking and improving your work (or room to over-run your time budget on one or two questions). The invigilator will tell you when you have had one hour, when you have 30 minutes remaining, and when you have five minutes remaining.

If you have an individual support plan with extra time for invigilated assessments (e.g. 25% extra time), you should plan a time management strategy that will fit with your time allowance.

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Section A (40 marks)

Section A will ask to you to make targeted descriptions and interpretations of two different thin sections. These questions are NOT asking for a full thin section description so the key to success is keeping your answers short and targeted – remember about 1 mark per minute. Before you start looking at the thin section, read the question in full and work out exactly what information is being asked for. Don't waste time adding extra details that don't directly address the question.

Part 1
Examine **thin section R1** and answer the following questions.
(This question can also be attempted using this Virtual Microscope thin section, focusing on Rotation 2 and the left-hand side of the thin section.)

1. What are the three principal silicate minerals in this thin section? [6]

2. Describe the key textural features of this thin section. Include a labelled sketch in your answer. [10]

3. Write two or three sentences interpreting the order of crystallization of the three principal silicate minerals. [4]

4. Suggest an appropriate name for this rock. [2]

For R1: *See practical 3 for more on R1*

1) Olivine, clinopyroxene (augite), plagioclase. (2 marks for each; 1 mark for a reasonable mis-identification such as opx.)

2) The texture is poikilitic (interstitial would also be acceptable). Euhedral olivines are enclosed by large anhedral crystals of plagioclase and/or clinopyroxene. The sketch should show rounded olivines fully enclosed by plagioclase and/or clinopyroxene; the sketch should include all three minerals to demonstrate that the textural associations of all three minerals have been noticed. Very good sketches may show the small oxides (label as magnetite, spinel or oxides) which can be found as inclusions within plagioclase and cpx, and occasionally within the olivines.

5 marks for sketch, 5 marks for labels/written description. Description doesn't have to be long or in full sentences but must be consistent with the sketch.

3) The coarse grain size requires a slow cooling rate, so this is a plutonic rock that could have formed in a sill or a solidifying magma chamber. The texture indicates that olivine (and oxides) crystallized first. The olivines are fairly euhedral so they grew free-floating in the magma. The plagioclase and clinopyroxene probably crystallized at about the same time, surrounding the olivines and filling the interstitial spaces. (4 marks; likely won't include all the detail given above.)

4) Coarse-grained rock with essential minerals plagioclase and clinopyroxene – this is a gabbro. The type mineral is olivine, so a full classification would be olivine gabbro. (2 marks for olivine gabbro; 1 for gabbro)

For Virtual Microscope:

1) Olivine, clinopyroxene (augite), plagioclase. (2 marks for each; 1 mark for a reasonable mis-identification such as opx.)

2) The texture in Rotation 2 is poikilitic (interstitial would also be acceptable). Euhedral olivines are enclosed by a large anhedral crystal of clinopyroxene. The texture of plagioclase is not clear from this rotation alone.
OR The texture on the left-hand side of the thin section is a cumulate texture. Euhedral tabular plagioclase laths have a distinct preferred orientation. Olivine and plagioclase are the cumulus phases; they form the main fabric of the rock. Clinopyroxene is the intercumulus phase; it fills the pore spaces and has an interstitial texture.

The sketch should include all three minerals to demonstrate that the textural associations of all three minerals have been noticed. Very good sketches may show the small oxides (label as magnetite, spinel or oxides) which can be found as inclusions within olivine, plag and cpx.

5 marks for sketch, 5 marks for labels/written description. Description doesn't have to be long or in full sentences but must be consistent with the sketch.

3) The coarse grain size requires a slow cooling rate, so this is a plutonic rock that could have formed in a sill or a solidifying magma chamber. Olivine and plagioclase have euhedral shapes indicating that they grew early, free-floating in a magma chamber. There is no textural indication of which began to crystallize first, so the simplest interpretation is that they crystallized at about the same time. The preferred orientation of plagioclase indicates that they settled through the magma. The clinopyroxene grew later, filling the pore spaces. (4 marks; likely won't include all the detail given above.)

4) Coarse-grained rock with essential minerals plagioclase and clinopyroxene – this is a gabbro. The type mineral is olivine, so a full classification would be olivine gabbro. (2 marks for olivine gabbro; 1 for gabbro)

Part 2
Examine **thin section SVH1** and answer the following questions.
(This question can also be attempted using this Virtual Microscope thin section; the phenocrysts are quite small so use the zoom function.)

5. What are the three principal phenocryst minerals in this thin section? *(Two phenocrysts if using Virtual Microscope example)* [6]

6. Describe and sketch two different disequilibrium textures shown in any of the phenocryst minerals. For each texture, write one or two sentences interpreting the igneous processes that could have formed it. [10]

7. Suggest a tectonic setting in which this magma could have formed. Justify your answer on the basis of your observations of the mineralogy of this thin section. [4]

For SVH1: *See practical 8 for more on SVH1*

5) Hornblende, plagioclase, orthopyroxene. (2 marks for each; 1 mark for a reasonable mis-identification.)

6) Any two of the following:
Plagioclase – Most crystals have oscillatory zoning, probably formed in response to changing magma composition due to magma mixing. Evidence for disequilibrium is the sieve textures within the oscillatory zoning, where the crystal has partially resorbed at its edges before continuing to grow.

Hornblende: (a) Crystals are often rounded, which forms when crystals partially resorb into the melt, thus evidence for disequilibrium. (b) Thin, dark, fine-grained reaction rims surround some crystals – evidence that the hormblende crystals spent time outside their stability field before the magma was erupted. The thicker the reaction rim, the longer the hornblende crystal spent outside the hornblende stability field.

Quartz (only in some thin sections): Rounded crystal shapes, indicating disequilibrium with surrounding melt.

5 marks per texture (only two required). Each texture must have a sketch and some words/labels of description; maximum three marks if sketch or description are omitted.

7) Must be a destructive plate margin (2 marks) which is the only tectonic setting where magmas contain enough water to crystallize hydrous minerals like hornblende (2 marks). Could also mention that complex disequilibrium textures are most common in subduction zone magmas. "Volcanic rock" is not a tectonic setting and will not be credited.

For Virtual Microscope:

5) Only two main phenocryst phases in this section: hornblende, plagioclase. But there are two types of hornblende: one brown, one green. The green hornblendes are rather small. (2 marks for each; 1 mark for a reasonable mis-identification e.g. chlorite instead of green amphibole.)

6) Any two of the following:
Plagioclase – Some phenocrysts have oscillatory zoning, probably formed in response to changing magma composition due to magma mixing. Evidence for disequilibrium is the sieve textures within the oscillatory zoning, where the crystal has partially resorbed at its edges before continuing to grow.

Hornblende – (a) Thin, dark, fine-grained reaction rims surround some crystals – evidence that the hormblende crystals spent time outside their stability field before the magma was erupted. The thicker the reaction rim, the longer the hornblende crystal spent outside the hornblende stability field. Some crystals have completely reacted out, leaving only black, fine-grained areas. Both brown and green hornblendes have reaction rims. (b) The green hornblendes are often subhedral to anhedral with rounded edges, indicating that chemical disequilibrium has led to resorption.

On the right side of the thin section is a coarse-grained area containing green amphibole and quartz; this is a xenolith that looks texturally unhappy with evidence of fine-grained reaction products around its edges.

5 marks per texture (only two required). Each texture must have a sketch and some words/labels of description; maximum three marks if sketch or description are omitted.

7) Must be a destructive plate margin (2 marks) which is the only tectonic setting where magmas contain enough water to crystallize hydrous minerals like hornblende (2 marks). Could also mention that complex disequilibrium textures are most common in subduction zone magmas. "Volcanic rock" is not a tectonic setting and will not be credited.

Section B (40 marks)

Section B will be a full thin section description. You will be asked to sketch, describe and interpret one thin section. The question will have the general format given below, and will include one or two targeted questions on important mineralogical or textural features. Read all the sub-questions before you start looking at the thin section, so that you know what features to look for and pay particular attention to, and so that you can be thinking about your interpretations as you prepare the sketch. The key to success is to take time and care over the details, especially the sketch – remember about 1 mark per minute.

8. Give a full petrographic description and interpretation of **thin section STR** or **thin section Re114**. Marks will be awarded as follows:

a) Provide **at least one** fully labelled sketch illustrating the rock's mineralogy and texture [15]

b) List the minerals in the rock, and estimate the rock's modal mineralogy [5]

c) *A targeted question asking you to interpret a named feature in the rock. Example:* Describe the shapes and sizes of the principal phenocryst phases. What information, if any, do the phenocrysts provide about the order of crystallization? [6]

d) *A targeted question asking you to interpret a second named feature in the rock. Example:* What information does the texture of this rock provide about the magma's thermal history (i.e., cooling rate)? [6]

e) Classify the rock using an appropriate petrological classification scheme, and write one sentence justifying your choice of classification scheme. [2]

f) Suggest a tectonic setting in which this rock may have formed. Justify your answer on the basis of your observations of the mineralogy and texture of this thin section. [6]

Most parts of this question using any igneous thin section on the Virtual Microscope, or any of the thin sections you have worked on in the practical classes. Make sure you practice describing a range of rocks: mafic, intermediate and silicic; volcanic and plutonic.

The solutions to Practical 4 contain an example of a full petrographic description and interpretation of a thin section.

Formative feedback on your own description and interpretation of sample STR or Re114 was given following Practical 6. The class-level feedback and solutions to Practical 6 contain some top tops on how to make sure you gain the easy marks, and common errors to avoid. It is worth consulting this carefully!

Section C (20 marks)

Section C is short-answer questions. You will be asked to classify rocks using geochemical or petrological information. You will be asked to demonstrate familiarity with symmetry, optics, and the structures of the rock-forming silicate mineral groups. The most common error in Section C is running out of time and not answering all the questions. Do not leave this section until the last possible moment – remember about 1 mark per minute. Make sure you are familiar with the contents of your Practical Reference Handout – this will save you time. Make sure you know the various classification schemes and how to use them. Add any extra annotations you think you may want.

9. The bulk composition of fine-grained igneous rocks A and B is given in Table 1. Classify both these rocks using an appropriate chemical classification scheme. [4]

Table 1: Bulk compositions of igneous rocks A and B

wt.%	**Rock A**	**Rock B**
SiO_2	57.4	39.7
TiO_2	0.6	2.8
Al_2O_3	19.5	11.4
Fe_2O_3	4.9	13.5
MnO	0.2	0.2
MgO	1.0	12.1
CaO	2.8	12.8
Na_2O	7.9	4.6
K_2O	5.4	2.0
P_2O_5	0.2	0.9

A: Phonolite (2) B: Nephelinite (2)
The question states that the rocks are fine-grained, so the rock names reported should be taken from the TAS diagram for volcanic rocks. Award 1 mark for each composition if the TAS diagram for plutonic rocks was used correctly.

10. The modal mineralogy and grain size of a very coarse-grained igneous rock C is given below. Classify this rock using an appropriate petrological classification scheme, and write one sentence justifying your choice of classification scheme. [3]

Rock C Average grain size 10 mm
Plagioclase, An_{30} 46%
Alkali feldspar 6%

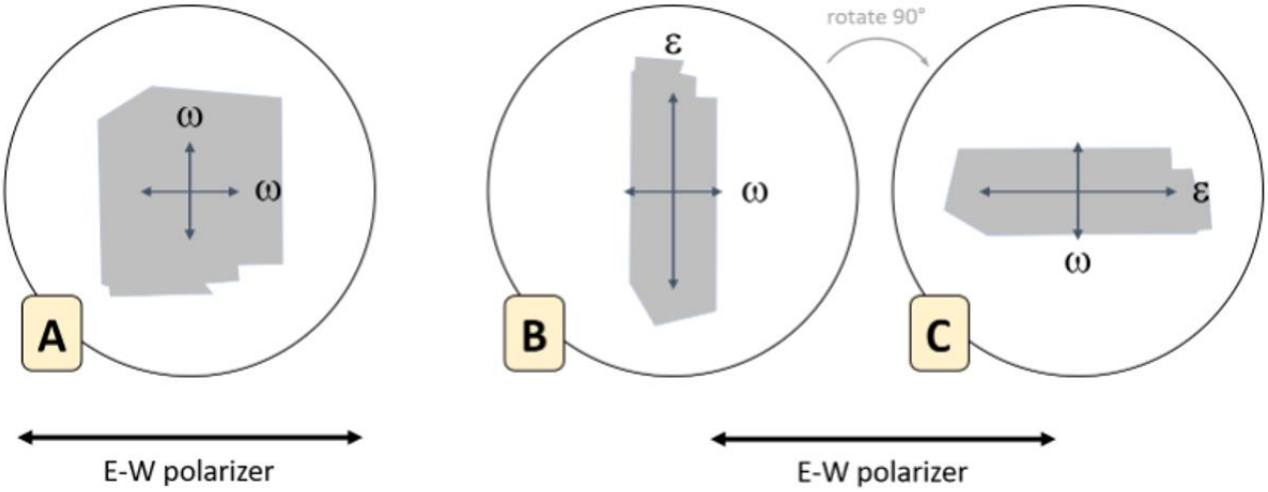
Quartz	27%
Muscovite	22%
Apatite	2%

Coarse-grained rock with Quartz + Alkali feldspar + Plagioclase all present, therefore best choice of detailed classification scheme is the QAP(F) diagram (1 mark). Credit any other sensible and justified choice of classification scheme.

Re-normalize Q, A and P to 100%, then use QAP(F) diagram. QAP classification is granodiorite. Rock name should include the type mineral before the root name: muscovite granodiorite. Apatite is a low-abundance accessory mineral, so not used in rock name.

Acceptable answers:
Muscovite granodiorite (2 marks)
Granodiorite *with no type mineral in the rock name* (1.5 mark)
Muscovite granite (1.5 mark)
Granite *with no type mineral in the rock name* (1 mark, only if justified by choice of classification scheme, e.g. the Gill simple mineralogical scheme)

11. The diagram below shows a schematic diagram of a mineral viewed under a petrological microscope in three different orientations. The arrows labelled ω and ε are the refractive indices. The mineral is pleochroic: refractive index ω is brown, and refractive index is ε yellow.



i) If this mineral is viewed under plane-polarized light, what will its colour be in each of orientations A, B and C? [3]
A: brown B: brown C: yellow

ii) If refractive index ω is nearly equal to the refractive index of the mounting medium, what is the relief of the mineral when viewed in orientation A? [2]
Low or very low relief: the crystal edges will appear thin and faint.

iii) If the mineral is viewed under cross-polarized light, what angle would you need to rotate orientation B in order to view the maximum interference colour? [2] 45°

12. The diagrams below show the crystallographic structure of a silicate mineral in two different orientations. In each diagram, the thin black lines show the unit cell.

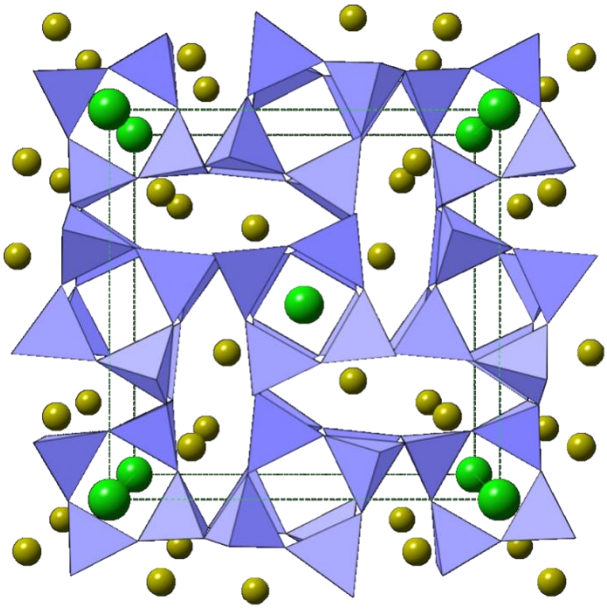
i) What type of silicate mineral is this? Select from: orthosilicate, single chain silicate, double chain silicate, sheet silicate, framework silicate. [2]
Framework silicate

ii) What type of unit cell is shown here? Select from: primitive (P), body-centred (I), face-centred (F), C face-centred (C). [1]
Body-centred

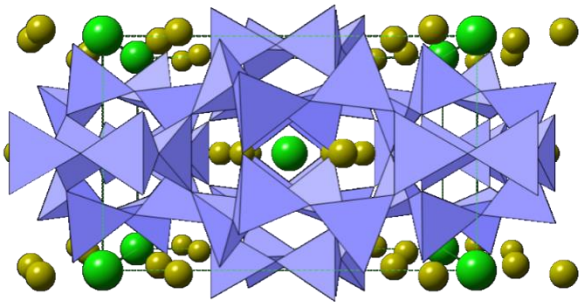
iii) What crystal system does this mineral belong to? Select from: cubic, trigonal, tetragonal, hexagonal, orthorhombic, monoclinic, triclinic. [1]
Tetragonal

iv) Is view A oriented looking down the crystallographic x, y or z axis? How do you know this? [2]
z axis (1), because the highest order of rotation is always parallel to z (1)

View A



View B



END OF TEST